Serhii Hirus

AI Automation & Backend Engineer · Remote, full-time

serhiihirus@gmail.com · Portfolio · LinkedIn · GitHub

SUMMARY

Automation engineer working at the intersection of Python, LLM APIs (Claude, OpenAI) and n8n. I design fault-tolerant systems: autonomous AI agents, RAG knowledge bases, CRM synchronization and media pipelines for product and engineering teams. Open to AI Automation Engineer, Python / AI Solutions Engineer and LLM Integration Developer roles.

SKILLS

- **AI agents and LLM pipelines:** Claude API (Anthropic), OpenAI API, LangGraph, tool calling, structured outputs (Pydantic), AST validation, retries and model fallbacks
- **RAG and knowledge bases:** Qdrant, pgvector, LangChain, hybrid search with reranking
- **Automation and integrations:** n8n (self-hosted and cloud), Make, webhooks, REST APIs, HubSpot, Pipedrive, Telegram, Slack
- **Backend and data:** Python (FastAPI, Asyncio, Pydantic), SQL, PostgreSQL, Playwright, JavaScript / Node.js
- **Media and speech:** Whisper, FFmpeg
- **DevOps:** Docker, Linux, Git / GitHub, CI/CD, Google Cloud (GCP)

SELECTED PROJECTS

AI Strategy Co-pilot (hirus.systems)

Python · Claude API · AST Validation · VectorBT · Pandas / NumPy · FastAPI

- Multi-turn domain-specific AI agent transforming vague trading hypotheses into strictly verified algorithmic specifications, eliminating hallucinations and unhandled risk boundaries.
- Socratic clarification loop on Claude API auditing missing risk boundaries (Stop-Loss levels, limit order boundaries, trade invalidation rules) and enforcing schema validation.
- Compiles verified Python code with static AST validation (disallowing unsafe imports/eval, vectorized signal logic, zero look-ahead bias).
- Result: 100% risk parameter coverage enforced before execution; reduces time from raw idea to backtest-ready specification from hours to < 30 seconds.

Automated Content Creator

Python · Claude API · Whisper · FFmpeg · n8n · PostgreSQL · Docker · Google Cloud (GCP)

- Built an autonomous pipeline that turns videos and articles into posts and subtitled clips for 6 platforms (LinkedIn, Telegram, X, Threads, Medium, Discord).
- Telegram bot as the control interface, Python microservices (parsing, Whisper, FFmpeg rendering) and a PostgreSQL scheduler.
- Claude API prompts and structured outputs adapt tone and style to each platform.
- Result: replaces 15+ hours of manual work per week (sourcing, clipping, subtitling, writing, posting) with one-click publishing or scheduled autoposting.

AI-copilot for CRM

Python · FastAPI · n8n · Apollo API · Whisper · HubSpot · Pipedrive · PostgreSQL

- Eliminated manual pre-call lead research and automated CRM post-processing after sales meetings.
- Engineered async service on FastAPI and n8n: inbound leads enriched via Apollo API, meeting audio transcribed with Whisper, BANT metrics extracted, AI deal scoring updated, and two-way sync with HubSpot/Pipedrive triggered.
- Result: lead response time reduced to 30 seconds; saves up to 12 hours weekly per sales rep on CRM updates and proposal drafting.

EDUCATION

Courses and certifications
- Google: Introduction to Generative AI, 2026.
- Coursera: Generative AI with Large Language Models, 2026.

LANGUAGES

English — B1 · Ukrainian — native · Russian — native